

Multiple Choice

1. **Answer:** A

Options: A) Synthesis of the blood clotting proteins; B) Prevention of bone loss; C) Regulation of gene expression and cell differentiation; D) Boosting immune system function

Note: The synthesis of blood clotting proteins is primarily a function of vitamin K, not vitamin A, which is involved in vision, immune function, and cell growth.

2. **Answer:** A

Options: A) Fat conducts electricity better than muscle; B) Fat is denser than muscle; C) Fat contains less water than muscle; D) Muscle weighs more than fat; E) Muscle contains less water than fat

Note: The correct answer is based on the fact that fat tissue has a higher water content and lower resistance to electrical flow compared to muscle, allowing electrical currents to travel more quickly through muscle.

3. **Answer:** B

Options: A) Exposure to extreme cold; B) Clogged blood vessels; C) Loss of collagen; D) Loss of skin elasticity; E) Decreased water intake

Note: Clogged blood vessels can lead to reduced blood flow and oxygen supply to muscles, resulting in neuronal damage and subsequent loss of muscle fibers due to the inability to stimulate and maintain muscle health effectively.

4. **Answer:** B

Options: A) Physiological; B) Behavioral; C) Biological; D) Anthropological; E) Environmental

Note: Research has primarily focused on the psychological and physiological mechanisms of social support's impact on health, while behavioral factors have been less thoroughly explored, leading to the conclusion that they have not received as much support in this context.

5. **Answer:** A

Options: A) Meta pneumo virus; B) Measles; C) Ebola; D) Nipah virus; E) Newcastle disease virus

Note: Metapneumovirus is known to be associated with severe respiratory illnesses, particularly in young children, and is a significant cause of croup due to its ability to induce inflammation of the upper airways.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The total drug dose is calculated by multiplying the person's weight (62 kg) by the drug dose per kilogram (15 mg/kg), resulting in 930 mg, which is equal to 0.93 grams.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the complete oxidation of one molecule of glucose through glycolysis, the citric acid cycle, and oxidative phosphorylation typically results in the production of approximately 32 molecules of ATP.

8. **Answer:** False

Options: A) True; B) False

Note: ABCA 1 primarily functions in the transport of cholesterol from cells to high-density lipoprotein (HDL) particles, rather than facilitating cholesterol uptake in the small intestine.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the net production of ATP via substrate-level phosphorylation in glycolysis is actually 2 ATP molecules from glucose and 3 ATP molecules from glycogen due to the differences in the starting substrates and their respective pathways.

10. **Answer:** False

Options: A) True; B) False

Note: Younger workers often benefit from job training as much, if not more than, their older counterparts due to their adaptability, familiarity with technology, and longer potential career lifespan to apply new skills.

Long Form Response

11. **Answer:** Polyomaviruses, a family of small, double-stranded DNA viruses, play a complex role in human health, primarily due to their ubiquitous presence and frequent asymptomatic infections in the general population. While certain polyomaviruses, such as JC virus and BK virus, have been linked to diseases like progressive multifocal leukoencephalopathy (PML) and renal complications in immunocompromised individuals, they are often considered non-pathogenic in healthy individuals. This non-pathogenicity can be attributed to the robust immune response elicited by the host, which effectively controls viral replication and prevents disease manifestation. As a result, most infections remain latent without causing any clinical symptoms, underscoring the importance of immune competence in mitigating the potential pathogenic effects of these viruses. Understanding this dynamic is crucial for public health, as it

highlights the need for vigilance in at-risk populations while recognizing the generally benign nature of polyomavirus infections in healthy individuals.

Note: correct because it accurately describes the dual role of polyomaviruses in human health, emphasizing their association with serious diseases in immunocompromised individuals while highlighting their common asymptomatic presence and non-pathogenic nature in healthy individuals.

12. **Answer:** The diagnostic criteria for osteopenia and osteoporosis in North America and Western Europe primarily hinge on bone mineral density (BMD) measurements obtained through dual-energy X-ray absorptiometry (DEXA). According to the World Health Organization (WHO), osteopenia is defined as a BMD that is 1 to 2.5 standard deviations below the mean for a young adult reference population, while osteoporosis is defined as a BMD that is 2.5 standard deviations or more below this mean. Age and gender significantly influence these assessments, as postmenopausal women, particularly those over age 80, often exhibit a decline in BMD due to hormonal changes. Interestingly, it has been observed that women in this age group may have higher-than-expected BMD for their age, which can complicate the diagnosis of osteopenia and osteoporosis, suggesting the need for careful interpretation of BMD data in the context of age-related bone health.

Note: correct because it accurately describes the WHO's definitions of osteopenia and osteoporosis based on specific BMD measurements, while also emphasizing the importance of age and gender as critical factors influencing these assessments.